

SANTA BARBARA BUOY 46054, CA, USA

WMO: 996470

Lat: 34.265N

Lon: 120.477W

Elev: 0

StdP: 14.70

Time Zone: -8.00 (NAP)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	48.4	49.5	N/A	N/A	N/A	N/A	N/A	N/A	36.6	51.0	35.0	51.1	18.4	320	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	3.9	68.5	N/A	66.8	N/A	65.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	11.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
34.5	32.6	31.0	46.1	71.9	2.6	2.8	44.2	73.9	42.7	75.5	41.2	77.1	39.3	79.1		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.6	55.9	54.7	54.3	53.3	54.2	55.8	57.9	58.6	59.6	60.2	58.7	56.0
	DBStd	3.52	3.24	2.79	2.84	2.05	2.04	2.09	2.70	2.48	3.20	3.36	2.82	2.75
	HDD50	4	1	1	1	1	0	0	0	0	0	0	0	0
	HDD65	3077	281	287	333	350	336	277	221	199	168	156	190	279
	CDD50	2418	186	134	133	101	129	174	245	266	288	315	261	186
	CDD65	14	0	0	0	0	0	1	1	5	6	1	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.9	14.3	15.6	17.2	18.4	19.6	19.1	18.0	18.9	17.6	15.2	14.2	14.4
Precipitation	PrecAvg	16.6	3.6	3.7	2.7	1.1	0.4	0.1	0.0	0.0	0.1	0.8	1.3	2.8
	PrecMax	32.6	16.3	15.7	12.6	4.3	2.4	0.7	0.3	0.1	0.4	4.0	4.5	9.3
	PrecMin	6.2	0.2	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2
	PrecStd	6.6	3.5	3.3	2.8	1.1	0.6	0.2	0.1	0.0	0.1	0.9	1.1	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.0	64.1	64.0	61.8	64.4	64.7	68.1	66.9	71.1	71.2	68.3	64.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	62.8	61.2	61.2	58.4	59.4	61.0	65.2	64.5	68.6	68.2	65.6	62.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	61.4	59.7	59.3	56.9	58.0	59.6	63.6	63.5	66.4	66.6	64.0	61.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	60.1	58.4	58.0	55.9	57.0	58.7	61.8	62.5	64.0	65.3	62.6	59.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	4.2	3.7	3.8	3.2	3.0	3.2	3.3	3.2	3.6	3.9	4.1	3.6
		MCDBR	6.3	5.2	6.0	4.7	4.8	4.8	3.9	3.7	5.1	5.6	6.4	4.3
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.306	0.315	0.332	0.352	0.361	0.347	0.370	0.351	0.347	0.349	0.312	0.312	
	taud	2.611	2.599	2.546	2.470	2.469	2.506	2.496	2.514	2.536	2.528	2.605	2.604	
	Ebn at Noon	290	298	300	296	293	296	288	293	291	281	285	280	
	Edh at Noon	24	27	31	35	35	34	34	33	31	29	24	23	
All-Sky Solar Radiation	RadAvg	974	1244	1669	2065	2216	2206	2063	1926	1719	1363	1075	872	
	RadStd	72	117	104	122	174	119	99	119	117	123	64	74	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+1.08	N/A	-2.68	+1.31	N/A	N/A	N/A	-245	+405	+137

Nomenclature: See separate page